

PART 1: LAWS OF EXPONENTS – LEVEL 1: EASY

Exercise 1

Simplify using positive exponents:

$$\frac{(x^{-4}y^{-2})^{-3}}{(x^3y^{-5})^{-2}}$$

Answer 1

1. Distribute powers:

$$= \frac{x^{12}y^6}{x^{-6}y^{10}}$$

2. Apply quotient rule:

$$= x^{12-(-6)}y^{6-10} = x^{18}y^{-4}$$

$$= \frac{x^{18}}{y^4}$$

Exercise 2

Simplify the product completely:

$$(a^{-2}b^3)^{-4} \cdot (a^5b^{-2})^{-1}$$

Answer 2

1. Distribute exponents:

$$= (a^8b^{-12}) \cdot (a^{-5}b^2)$$

2. Add powers of like bases:

$$= a^{8-5}b^{-12+2} = a^3b^{-10}$$

$$= \frac{a^3}{b^{10}}$$

Exercise 3

Simplify using positive exponents:

$$\frac{(u^3v^{-4})^{-2}}{(u^{-1}v^{-2})^4}$$

Answer 3

1. Apply power of a power:

$$= \frac{u^{-6}v^8}{u^{-4}v^{-8}}$$

2. Subtract exponents:

$$= u^{-6-(-4)}v^{8-(-8)} = u^{-2}v^{16}$$

$$= \frac{v^{16}}{u^2}$$

PART 1: LAWS OF EXPONENTS – LEVEL 2: MEDIUM

Exercise 4

Simplify fully with positive exponents:

$$\frac{(x^{-5}y^4)^{-2}}{(x^2y^{-3})^{-3}}$$

Answer 4

1. Expand power terms:

$$= \frac{x^{10}y^{-8}}{x^{-6}y^9}$$

2. Subtract exponents:

$$= x^{10-(-6)}y^{-8-9} = x^{16}y^{-17}$$

$$= \frac{x^{16}}{y^{17}}$$

Exercise 5

Simplify the fractional exponent quotient:

$$\frac{(a^{-4}b^6)^{\frac{3}{2}}}{(a^2b^{-8})^{\frac{1}{2}}}$$

Answer 5

1. Multiply inner by outer powers:

$$= \frac{a^{-6}b^9}{a^1b^{-4}}$$

2. Combine exponents:

$$= a^{-6-1}b^{9-(-4)} = a^{-7}b^{13}$$

$$= \frac{b^{13}}{a^7}$$

Exercise 6

Simplify the product of fractions:

$$\left(\frac{x^{-3}y^{\frac{2}{3}}}{x^2y^{-\frac{4}{3}}}\right)^{-3} \times \left(\frac{x^4y^{-2}}{x^{-1}y^3}\right)^{\frac{1}{2}}$$

Answer 6

1. Simplify inside brackets:

$$= (x^{-5}y^2)^{-3} \times (x^5y^{-5})^{\frac{1}{2}}$$

2. Apply outer exponents:

$$= (x^{15}y^{-6}) \cdot (x^{\frac{5}{2}}y^{-\frac{5}{2}})$$

3. Add exponents:

$$= \frac{x^{\frac{35}{2}}}{y^{\frac{17}{2}}} = \frac{x^{17}\sqrt{x}}{y^8\sqrt{y}}$$

Exercise 7

Evaluate numerical bases and simplify:

$$\frac{(8x^{-3}y^6)^{-\frac{4}{3}}}{(32x^5y^{-10})^{\frac{2}{5}}}$$

Answer 7

1. Evaluate numerical powers:

$$8^{-\frac{4}{3}} = (2^3)^{-\frac{4}{3}} = \frac{1}{16}, \quad 32^{\frac{2}{5}} = 4$$

2. Distribute powers to variables:

$$= \frac{\frac{1}{16}x^4y^{-8}}{4x^2y^{-4}} = \frac{1}{64}x^{4-2}y^{-8-(-4)}$$

$$= \frac{x^2}{64y^4}$$

PART 1: LAWS OF EXPONENTS – LEVEL 3: HARD

Exercise 8

Simplify the nested power structure:

$$\left[\frac{(m^{-3}n^2)^{-3} \cdot (m^2n^{-4})^2}{(m^{-4}n^3)^{-2}} \right]^{-\frac{2}{3}}$$

Answer 8

1. Distribute powers inside brackets:

$$= \left[\frac{(m^9n^{-6}) \cdot (m^4n^{-8})}{m^8n^{-6}} \right]^{-\frac{2}{3}} = \left[\frac{m^{13}n^{-14}}{m^8n^{-6}} \right]^{-\frac{2}{3}}$$

2. Simplify fraction inside:

$$m^5n^{-8}$$

3. Apply power $-\frac{2}{3}$:

$$= m^{-\frac{10}{3}} n^{\frac{16}{3}} = \frac{n^5 \sqrt[3]{n}}{m^3 \sqrt[3]{m}}$$

$$= \frac{n^5 \sqrt[3]{n}}{m^3 \sqrt[3]{m}}$$

Exercise 9

Simplify the complex 3-variable expression:

$$\frac{\left(\frac{x^{-\frac{1}{2}}y^{\frac{3}{4}}}{z^{-2}} \right)^{-4} \div \left(\frac{x^2z^{-1}}{y^{-\frac{1}{2}}} \right)^3}{\left(x^{-3}y^2z^{-\frac{1}{2}} \right)^{-2}}$$

Answer 9

1. Numerator term 1:

$$x^2y^{-3}z^{-8}$$

2. Numerator term 2:

$$x^6y^{\frac{3}{2}}z^{-3}$$

3. Quotient in numerator:

$$x^{-4}y^{-\frac{9}{2}}z^{-5}$$

4. Divide by denominator $(x^6y^{-4}z^1)$:

$$= x^{-10}y^{-\frac{1}{2}}z^{-6}$$

$$= \frac{1}{x^{10}z^6\sqrt{y}}$$

Exercise 10

Evaluate coefficients and simplify:

$$\frac{(125x^{-9}y^{15}z^{-3})^{-\frac{1}{3}} \cdot (16x^{-8}y^{12}z^4)^{\frac{3}{4}}}{(4x^{-4}y^8z^{-6})^{\frac{1}{2}}}$$

Answer 10

1. Numerical terms:

$$125^{-\frac{1}{3}} = \frac{1}{5}, \quad 16^{\frac{3}{4}} = 8, \quad 4^{\frac{1}{2}} = 2$$

2. Variable powers in numerator:

$$\left(\frac{1}{5}x^3y^{-5}z^1 \right) \cdot (8x^{-6}y^9z^3) = \frac{8}{5}x^{-3}y^4z^4$$

3. Divide by denominator $(2x^{-2}y^4z^{-3})$:

$$= \frac{4z^7}{5x}$$